

Test 1 (Class 9 ICSE)

Time: 1 hour 30 minutes

Marks : 60

“Programming sharpens the mind to perceive the universe’s order—hinting that reality itself may be a simulation, crafted like code”

MCQ.

1. Mnemonics are used in _____
 - a. Machine language
 - b. Assembly Language
 - c. Procedure oriented languages
 - d. Object Oriented languages
2. A structure-oriented language is
 - a. C
 - b. Java
 - c. Assembly
 - d. Pascal
3. The code we write is
 - a. Source code
 - b. interpreted code
 - c. Object code
 - d. Compiled code
4. The code computer receives after conversion is
 - a. Source code
 - b. interpreted code
 - c. Object code
 - d. Compiled code
5. Extension of JAVA Source code is
 - a. .Source
 - b. .java
 - c. .class
 - d. .program
6. Extension of JAVA byte code
 - a. .source
 - b. .java
 - c. .class
 - d. .program
7. Three properties of an object are
 - a. Identity, attribute, behavior
 - b. shape, identity, type
 - c. type, behavior, attribute
 - d. shape, attribute, behavior
8. Scanner sc=new Scanner(System.in) , here 'sc' is
 - a. Object
 - b. reference variable
 - c. address variable
 - d. primitive variable
9. Default package in JAVA is
 - a. java.util
 - b. java.lang
 - c. java.net
 - d. java.applet

10. Polymorphism is
- Defining multiple functions with different parameters
 - Combining multiple units into one
 - Taking properties of another class
 - Showing necessary details and hiding background details
11. To take character as input, we can write (assuming sc is the Scanner reference)
- sc.next()
 - sc.nextLine()
 - sc.nextCharAt(0)
 - sc.next().charAt(0)
12. To terminate the whole program _____ is used
- break
 - System.exit(0)
 - end
 - ;
13. Math.round() returns value in
- long
 - float
 - int
 - double
14. Symbolic constant is declared using _____ keyword
- final
 - finally
 - finalize
 - const
15. Is this code valid? `-> char ch = 'A' + 'B';`
- Yes
 - No
 - Depends on Computer's mood
 - Depends on whether it's a rainy day or not
16. Data type of k if `k = 10 + 1.5 + 'A' + "Hello World"` would be
- Double
 - long
 - char
 - String
17. Absence of break causes fall through in
- If-else
 - ternary
 - switch case
 - if-else if -ladder
18. The data types which are not allowed in switch parameter and case labels (multiple answers)
- int
 - long
 - String
 - double
19. Size difference between ASCII and Extended ASCII character is
- 1 byte
 - 2 bytes
 - 3 bytes
 - 4 bytes
20. `int k = 5 - '0'; k = ?`
- 5
 - error
 - 0
 - NaN

SAQ & Snippets:

1. String s= 2 + 5 + "Hello" + 2 + 5
System.out.println(s);
2. int x=5, y=10;
x*= x++ + ++x - --y % (x*y)
System.out.println(x)
System.out.println(x==y);
3. generate a random number between 25 and 50 (both included)
4. what range would the following expression return : (int) (Math.random() *13) + 4
5. char ch1= 'A' + 32;
char ch2= 'a' - 32;
int r= 10;
System.out.println(ch1 + "" + ch2 + "" + (ch1+ch2+r));
6. System.out.println(Math.ceil(2.3) + Math.ceil(-2.3));
7. Convert the following if-else to ternary
if(x==y and x== z) y=x;
else if(x-y==x) y=0;
else y=y;
8. Convert the following if-else to switch-case
if((a==1 || a==2) && b==3) System.out.println("Yo");
else System.out.println("Nothing");
9. Check if a number is a multiple of 3 and ends with 1 or not using switch case
10. Write the following expression using JAVA 🐼

$$S = \left| \frac{\sqrt[5]{(x-y)^k}}{\sqrt{x^y - y^{10x}}} \right|$$

11. Write the mathematical expression which is depicted by the following java expression
double x= Math.pow(Math.pow(a, (c-k)), (1/k))
12. double k= Math.max(10e⁻⁵, 10e⁻⁶);
find k.

13. find output if (i) ch= 'A' (ii) ch= 'B'

```
switch(ch){  
case 'A': System.out.println("ONE");  
  
case 'C' System.out.println("Two");  
    break;  
default: System.out.println("Three");  
}
```

14. Find greatest between three numbers using ternary (You can't use Logical Operators)

15. What are comments in JAVA ? give example of each

16. What is the difference between $a = a * 10 + b$ and $a * = 10 + b$

17. Find output :

```
int x=2, y=15; double s=0;
```

```
if(x>y){  
    s= x*y;  
}else{  
    s= y/ (double) x;  
}  
System.out.println(s);
```

18. Write down the syntax for creating object

19. $!((2 > 3 \parallel 3 < 4) \&\& 5 != 3)$ find the returning boolean value

20. Why this won't generate a random integer between 1 and 6

```
int x= (int)Math.random() * 6 + 1;
```

Bonus Question: 10 Marks EXTRA AAAA 🏠 🧐 🤖

```
int x = (int)(Math.random() * ((4 * n + 1) - (2 * m - 3) + 1)) + (2 * m - 3);
```

Find the **range of possible values of x**.

Subhasis